VIDEO 9:

# HEXADECIMAL → DECIMAL USING THE POSITIONAL METHOD

Number System Made Easy Series



# UNDERSTANDING POWERS OF 16

Hexadecimal is **base 16**.

Each position represents a power of 16 starting from the right.

Multiply **each digit** by its positional power of 16.

## POWERS OF 16

| | | |
|---|---|---|
| $16^0$ | → | 1 |
| $16^1$ | → | 16 |
| $16^2$ | → | 256 |

## HEX DIGIT VALUES

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| 10 | 11 | 12 | 13 | 14 | 15 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# WORKED EXAMPLE 1

$1A_{16} = ?_{10}$

| | | | |
|---|---|---|---|
| 0<br>1 | $1 \times 16^1$ | = | $1_{16}$ |
| 0<br>2 | $1 \times 16^0$ | = | $1_0$ |
| 0<br>3 | $1 + 1$<br>$6 \quad 0$ | = | $2_6$ |

**THEREFORE**

$$1A_{16} = 26_{10}$$

# WORKED EXAMPLE

2

2 1

F represents 15. 16

F

$2 \times 16^1 \rightarrow 32$

$1 \times 16^0 \rightarrow 15$

$32 + 15 = 47$

$2F_{16} = 47_{10}$

# PRACTICE QUESTION

Your turn!

Convert $3B_{16}$ to decimal.

Try it in your head before we reveal the answer.



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**LET'S CHECK**

B = 11 → 3 × 16 = 48 → 11 × 1 = 11 → 48 + 11 = 59

**Answer:** $59_{10}$

# RECAP

Convert hexadecimal to decimal using **powers of 16**

Remember hex digits **A=10 to F=15**

**Multiply** each digit by its positional value

**Add** all products for final decimal value

NEXT UP →

**Next video:**
Octal and Hexadecimal Conversion




## THANKS FOR WATCHING!